

Chapter 8

Internal Forces

One of the fundamental assumptions we make in statics is that bodies are *rigid*, that is, they do not deform, bend, or change shape. While we know that this assumption is not true for real materials, we are building the analytical tools necessary to analyze deformation. In this chapter you will learn to compute the forces and moments inside a object which hold it together as it supports its own weight and any applied loads.

The chapter begins with a discussion of internal forces and moments and defines a new sign conventions especially for them. Next we will determine internal forces at a specific point within a rigid body. Finally, we develop three techniques to find internal forces at every point throughout a beam. Note that we use the words **internal forces** when we are referring to both “internal forces and internal bending moments.”

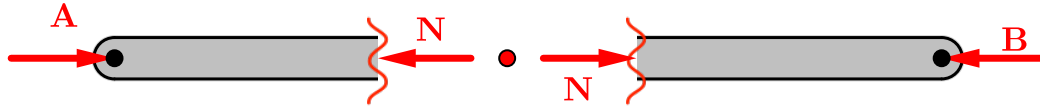
Determination of the internal forces is the first step in the engineering design of a structure. A properly designed structure must safely support all expected external loads, including live loads, dead loads, wind and earthquake loads. External loads produce internal forces, which in turn creates stresses, strains, and deformations in the structure. In a successful design, the shape, size, and material must all be carefully chosen to limit them to safe values. You are advised to pay attention, and master this topic.

8.1 Internal Forces

In [Subsection 3.3.3](#) you were introduced to axial loadings, which were either tension or compression, or possibly zero. This section will explain two other internal forces found in two-dimensional systems, the **internalshear** and **internalbending moment**.

Internal forces are present at every point within a rigid body, but they always occur in equal-and-opposite pairs which cancel each other out, so they’re not obvious. They’re there however, and when an object is cut (in your imagination) into two parts the internal forces become visible and can be determined.

You are familiar with straight, two-force members which only exist in equilibrium if equal and opposite forces act on either end. Now imagine that we cut the member at some point along its length. To maintain equilibrium, forces must exist at the cut, equal and opposite to the external forces. These forces are internal forces.



[Live Link](#)

Figure 8.1.1 Internal forces in a straight two-force member.

Now let's examine the two-force member shown in [Figure 8.1.2](#). This time, the member is L shaped, not straight, but the external forces must still share the same line of action to maintain equilibrium. If you cut across the object, you will obtain two rigid bodies which must also be in equilibrium. However, adding an equal and opposite horizontal force at the cut won't produce static equilibrium because the two forces form a couple which causes the piece to rotate. This means that something is missing!

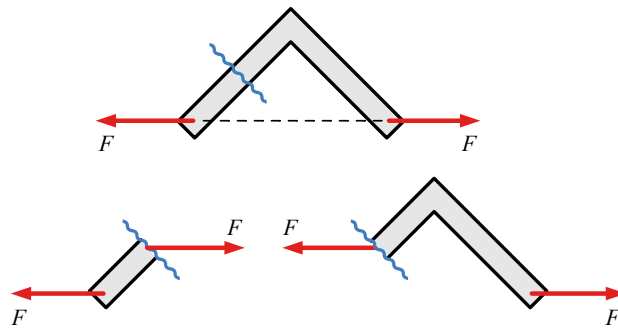


Figure 8.1.2 A horizontal force alone does not create equilibrium.

Two-dimensional rigid bodies have three degrees-of-freedom and require three equilibrium equations to satisfy static equilibrium in order to prevent translation in the x direction, the y direction, and to prevent rotation about the z axis.

Assuming the material is rigid, the connection between the two halves must resist both translation and rotation, so we can model this connection as a fixed support and replace the removed half of the link with a force reaction *and* a couple-moment reaction as shown in the free-body diagrams of [Figure 8.1.3](#). This internal loading is actually a simplification of a more complex loading distributed across the section plane. The couple $\mathbf{M}$ represents the net rotational effect of the force system on the surface of the cut.

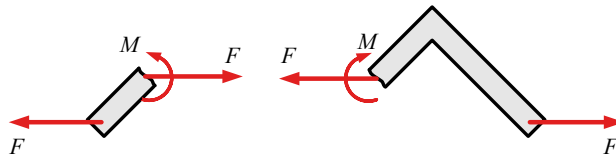


Figure 8.1.3 The internal forces are represented as an equal and opposite force $\mathbf{F}$ and a bending moment $\mathbf{M}$

The horizontal force can also be resolved into orthogonal components parallel and perpendicular to the cut. These components have special names in the context of internal forces.

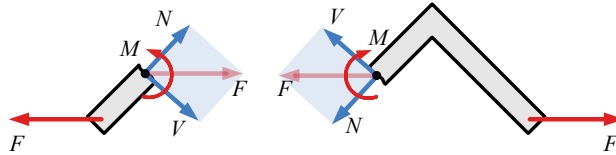


Figure 8.1.4 The internal forces are represented as a normal force $\mathbf{N}$, a shear force $\mathbf{V}$, and bending moment $\mathbf{M}$

The internal force component perpendicular to the cut is called the **normal force**. This is the same internal tension or compression force that we assumed to be the only significant internal load for trusses. If the object has an axis, and the cut is perpendicular to it, the normal force may also be properly called an **axial force**.

The internal force component parallel to the cut is called the **shear force**. The word shear refers to the shearing that occurs between adjacent planes due to this force. You can get a feel for shearing adjacent planes by sliding two pieces of paper together.

The internal couple-moment is called the **bending moment** because it tends to bend the material by rotating the cut surface.

The shear force is often simply referred to as **shear**, and the bending moment as **moment**; together with the normal or axial force the three together are referred to as the “internal forces”. The symbol $\mathbf{V}$ is commonly chosen for the shear force, and $\mathbf{A}$, $\mathbf{P}$ or $\mathbf{N}$ for the normal force and $\mathbf{M}$ for the bending moment.

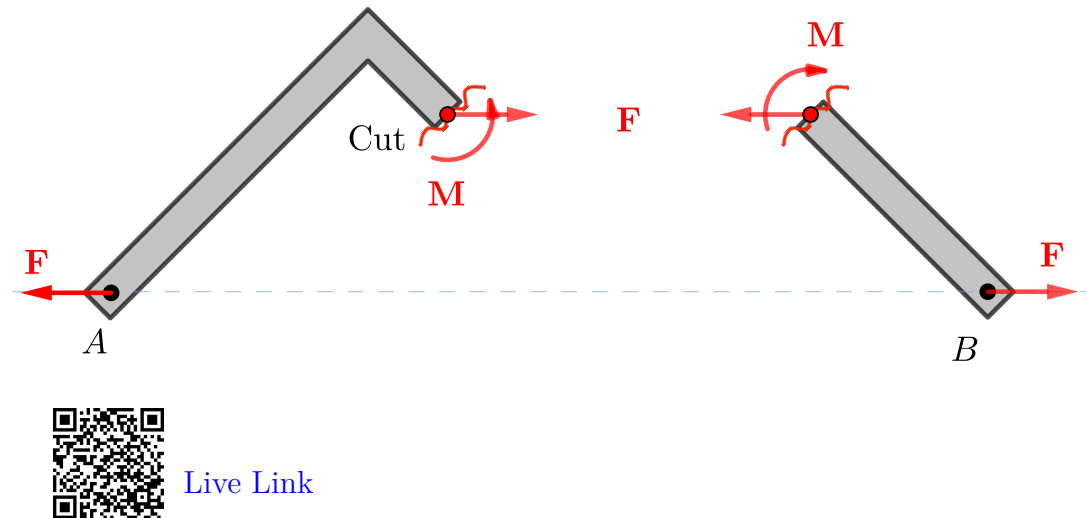


Figure 8.1.5 Internal Loading in a L shaped member.

Thinking Deeper 8.1.6 Deformation.

The controlling design parameter for most engineering systems is deformation. Thankfully, due to a property called elasticity, most materials will bend, stretch, and compress, long before they ultimately break. For example, when designing the floor in a new building, the floor is often limited to deflecting less than the length of the span in inches, divided by 360. Any more deformation than this would be considered disconcerting to the building residents and also start damaging surface materials like drywall. For example, for a 20 ft span, the deflection would need to be less than

$$\delta = \frac{20 \text{ ft} \cdot \frac{12 \text{ in}}{1 \text{ ft}}}{360} = 0.667 \text{ in.}$$

To meet this deformation limit, we need to consider the magnitude and location of applied loads, the size and shape of the floor beams, and the material the floor beams are made from. As deflection is an internal property of the flooring materials, the first step is to determine the internal forces that arise from the externally applied loads, using the methods of this chapter.

8.2 Sign Conventions

When talking about internal forces our standard sign convention for forces and moments is not good enough. We can't, for instance, just call a vertical shear force positive if it points up and negative if it points down, because internal

forces always occur in pairs so at any given point a shear force is both up *and* down. The direction of the internal force at a point depends on which side of the cut you're looking at.

So to define the state of internal forces at a point we need a better sign convention. Although the choice is somewhat arbitrary, agreeing on a *standard* sign convention allows us to have consistency across our calculations and to communicate the internal state clearly to others. The standard sign conventions defined here are used for internal loadings at a point and also for the shear and bending moment diagrams which are discussed in [Section 8.4](#).

Be aware that although this new sign convention applies to internal forces, it doesn't change the sign convention for the equations of equilibrium at all, so you will continue to solve them in the same way you always have.

The standard sign convention used for shear force, normal force, and bending moment is shown below.

- *Positive Shear.*



Positive shear forces tend to skew an object as shown, i.e. positive shear forces push down when looking from the right, and up when looking from the left.

- *Positive Normal Force.*



Positive normal forces tend to stretch the object.

- *Positive Bending Moment.*



Positive bending moments tend to deform the object with an upward curvature.

Question 8.2.1

We have defined positive internal forces by looking at the “front” side of the object. Would the results change if you walked around the object and analyzed it from the other side?

8.3 Internal Forces at a Point

This section covers the procedure to compute the internal normal force, shear force, and bending moment at a designated point in a multi-force rigid body.

Consider the frame shown in Figure 8.3.1 consisting of two-force members GD and BE , and multi-force members AD and CF . Since no information is provided, we can assume that the components have negligible weight. The internal loading within the two-force members is purely axial, but the multi-force members will be subject to the complete set of shear force, normal force, and bending moment.

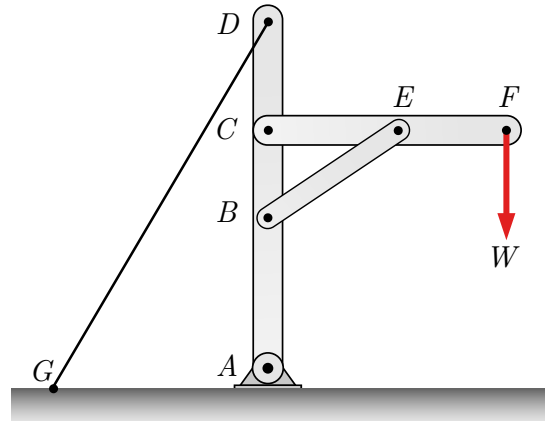
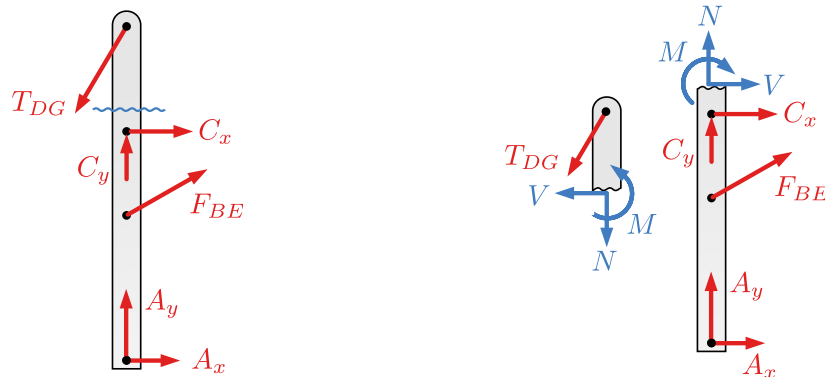


Figure 8.3.1 A frame supporting a load at F .

To find the internal forces at a specified point within one of the members, we make an imaginary cut there.



(a) Wavy line indicates the location of the imaginary cut.

(b) Internal forces are exposed by the cut.

Figure 8.3.2

Figure 8.3.2(a) shows the free-body diagram of member AD , with a proposed cut between points D and C . The free-body diagram is shown with reactions for pinned connections at A and C and forces from the two-force members at locations B and D .

Figure 8.3.2(b) shows the free-body diagrams for each half, with the three internal forces exposed and labeled V , N , and M . This is analogous to the Method of Sections technique of Chapter 6.

Note that the internal forces at the cut are drawn in the positive direction according to the [sign conventions](#) for internal forces, that they act in opposite direction either side of the cut, and they cancel out if the object is put back together.

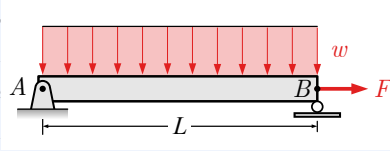
Either free-body diagram can be used to solve for the internal forces, so it is wise to choose the easier one. Recognizing which one is easier takes practice,

but look for the piece with more known and fewer unknown values.

This technique can be used to find the internal forces at any point within any object. In the examples below we will find the internal loadings at a specific point in load carrying beams.

Example 8.3.3 Internal forces in a simply supported beam.

A beam of length L is supported by a pin at A and a roller at B and is subjected to a horizontal force F applied to point B and a uniformly distributed load over its entire length. The intensity of the distributed load is w with units of [force/length].

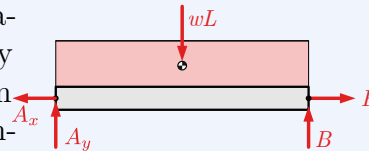


Find the internal forces at the midpoint of the beam.

Solution.

1. Find the external reactions.

Begin by drawing a free-body diagram of the entire beam, simplified by replacing the distributed load w with an equivalent concentrated load at the centroid of the rectangle.



The magnitude of the equivalent load W is equal to the “area” under the rectangular loading curve.

$$W = w(L)$$

Then apply and simplify the equations of equilibrium to find the external reactions at A and B .

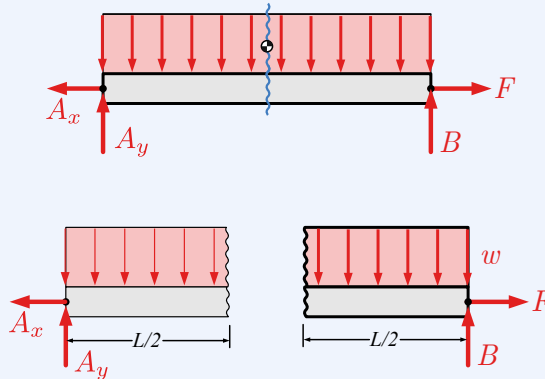
$$\begin{aligned}\Sigma M_A &= 0 \\ -(wL)(L/2) + (B)L &= 0 \\ B &= wL/2\end{aligned}$$

$$\begin{aligned}\Sigma F_x &= 0 \\ -A_x + F &= 0 \\ A_x &= F\end{aligned}$$

$$\begin{aligned}\Sigma F_y &= 0 \\ A_y - wL + B &= 0 \\ A_y &= wL - wL/2 \\ &= wL/2\end{aligned}$$

2. *Cut the beam.*

Cut the beam at the point of interest and separate the beam into two sections. Notice that as the beam is cut in two, the distributed load w is cut as well. Each of these distributed load halves will support equivalent point loads of $wL/2$ acting through the centroid of each cut half.

3. *Add the internal forces.*

At each cut, a shear force, a normal force, and a bending moment will be exposed, and these need to be included on the free-body diagram.

At this point, we don't know the actual directions of the internal forces, but we do know that they act in opposite directions. We will assume that they act in the positive sense as defined by the [standard sign convention](#).

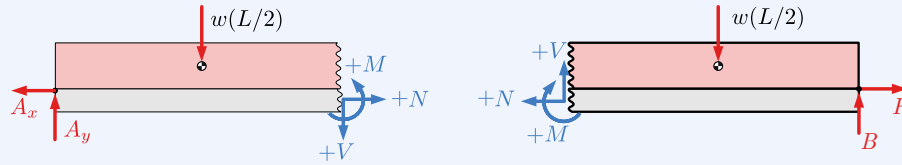
Axial forces are positive in tension and act in opposite directions on the two halves of the cut beam.

Positive shear forces act down when looking at the cut from the right, and up when looking at the cut from the left. An alternate definition of positive shears is that the positive shears cause clockwise rotation. This definition is useful if you are dealing with a vertical column instead of a horizontal beam.

Bending moments are positive when the moment tends to bend the beam into a smiling U-shape. Negative moments bend the beam into a frowning shape.

For vertical columns, positive bending moments bend a beam into a C shape and negative into a backward C-shape.

The final free-body diagrams look like this.



Horizontal beams should always have assumed internal loadings in these directions at the cut, indicating that you have assumed *positive* shear, *positive* normal force and *positive* bending moments at that point.

4. *Solve for the internal forces.*

You may use either FBD to find the internal forces using the techniques you have already learned. So, with a standard xy coordinate system, forces to the right or up are positive when summing forces and counter-clockwise moments are positive when summing moments.

Using the left free-body diagram and substituting in the reactions, we get:

$$\Sigma F_x = 0$$

$$-A_x + N = 0$$

$$N = A_x$$

$$\Sigma F_y = 0$$

$$A_y - wL/2 - V = 0$$

$$V = wL/2 - wL/2$$

$$V = 0$$

$$\Sigma M_{\text{cut}} = 0$$

$$(wL/2)(L/4) - (A_y)(L/2) + M = 0$$

$$M = -wL^2/8 + wL^2/4$$

$$M = wL^2/8$$

Using the right side free-body diagram we get:

$$\Sigma F_x = 0$$

$$-N + F = 0$$

$$N = F$$

$$\Sigma F_y = 0$$

$$V - wL/2 + B_y = 0$$

$$V = wL/2 - B_y$$

$$V = wL/2 - wL/2$$

$$V = 0$$

$$\Sigma M_{\text{cut}} = 0$$

$$-M - (L/4)(wL/2) + (L/2)(B_y) = 0$$

$$M = -wL^2/8 + wL^2/4$$

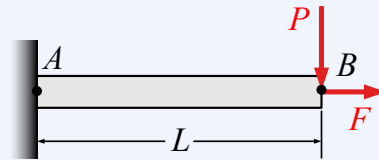
$$M = wL^2/8$$

Regardless of which side is chosen, we get the same results for the internal forces at the chosen point.

When you solve for the internal forces, the results can be either positive, negative, or sometimes zero. Negative values indicate that the actual direction of the load is opposite to the assumed direction. Since we assumed all three internal forces were positive as defined by the [standard sign convention](#), a negative answer means that the load actually acts in the opposite direction to the vector shown on the free-body diagram.

Example 8.3.4 Internal forces in a cantilever beam.

Consider a cantilever beam which is supported by a fixed connection at A , and loaded by a vertical force P and horizontal force F at the free end B . Determine the internal forces at a point a distance a from the left end.

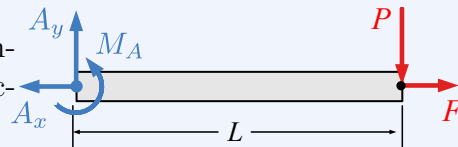


Hint. If you think ahead, you may not need to find the reactions at A .

Solution.

1. Determine the reactions.

Draw an FBD of the entire, uncut beam and determine the reactions.



Notice that only the applied loads and support reactions are included on this uncut beam FBD. The internal forces are only exposed and shown on a FBD after the beam is cut.

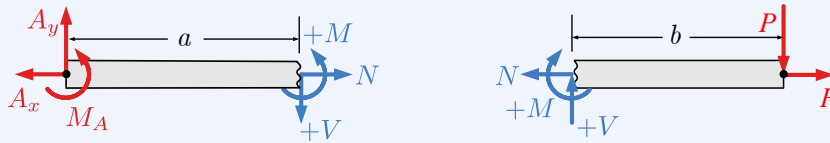
Use this free-body diagram and the equations of equilibrium to de-

termine the external reaction forces.

$$\begin{aligned}\Sigma F_x = 0 &\implies A_x = F \\ \Sigma F_y = 0 &\implies A_y = P \\ \Sigma M_A = 0 &\implies M_A = PL\end{aligned}$$

2. Section the beam.

Take a cut at the point of interest and draw a FBD of either or both parts. Try to choose the simpler free-body diagram. If one side has no external reactions, then you can skip the previous step if you choose that side.



The free-body diagrams of both portions have been drawn with the internal forces and moments drawn in the positive direction defined by the [standard sign convention](#).

The axial force is shown in tension on both parts. This force has been named N so its name doesn't conflict with the forces at point A .

The shear force V is positive when the shear is down on the right face of the cut and up on the left face.

The bending moment M is positive if the bending direction would tend to bend the beam into a concave upward curve.

Always assume that the unknown internal forces act in the positive direction as defined by the standard sign convention.

3. Solve for the internal forces.

Selecting the right hand diagram and solving for the unknown internal forces gives:

$$\begin{aligned}\Sigma F_x = 0 &\implies N = F && \text{from before } F = A_x \\ \Sigma F_y = 0 &\implies V = P && \text{from before } P = A_y \\ \Sigma M_{\text{cut}} = 0 &\implies M = -Pb = -P(L - a) && \text{since } a + b = L\end{aligned}$$

. Solving the other free-body diagram would produce the same results

Once you have found the reactions and drawn a free-body diagram of the simpler portion with the normal force, shear force, and bending moment assumed positive, you then solve for the unknown values and signs just like any other equilibrium problem.

This workflow typically includes:

- Establishing a horizontal x and vertical y coordinate system.
- Taking a cut at the point of interest.
- Assuming that the internal forces act in the positive direction and drawing a free-body diagram accordingly
- Using $\Sigma F_x = 0$, $\Sigma F_y = 0$, and $\Sigma M_z = 0$ to solve for the three unknown internal forces.

The shear force V , normal force N , and bending moment M are scalar components and they may be positive, zero, or negative depending on the applied loads. The signs of the scalar components together with the sign convention for internal forces establish the actual directions of the shear force, normal force and bending moment vectors.

8.3.1 Interactive Internal Forces

The internal forces and bending moments inside a beam depend on the load that the beam is supporting and differ from point to point. This simply supported beam supports a uniformly varying load. The interactive traces out the value of the shear and bending moment as you move point C . Can you deduce the relation between the triangular loading and the value of the shear and bending moment?

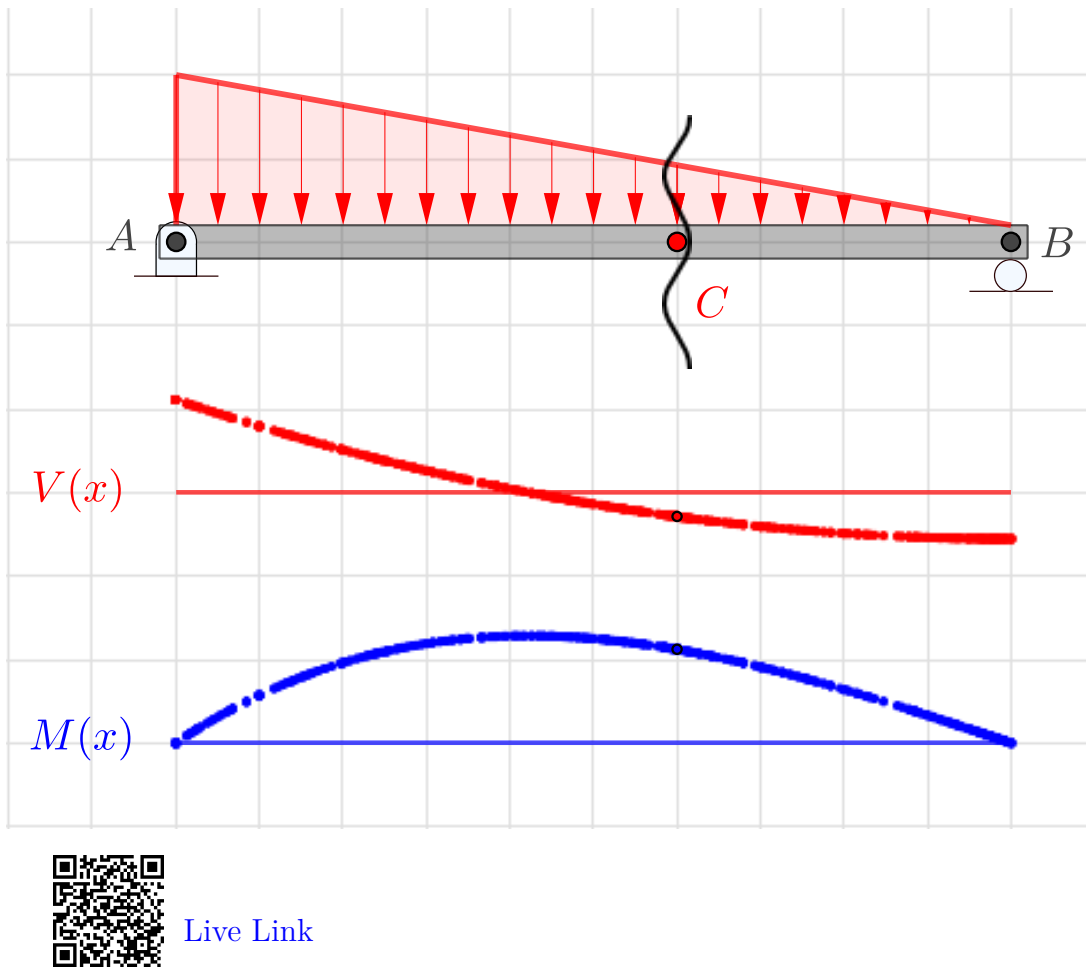


Figure 8.3.5 Internal forces in a beam with a uniformly varying load.

8.4 Shear and Bending Moment Diagrams

Shear and moment diagrams are graphs which plot the internal shear and bending moment against the length of the beam.

They allow us to *see* the internal forces throughout the entire beam so that we can optimize the design to prevent failures and reduce the overall weight and cost of the structure. A beam is most likely to fail at points where shear and moment are greatest and these diagrams help us find these points.

The previous section presented a method to find the internal forces at a *single* point, which is useful; but to find the shear and moment at *every* point we need a more powerful approach. The following sections discuss three different methods to produce shear and bending moment diagrams, and concludes with a comparison of the advantages and disadvantages of each approach.

Beams can be supported in a variety of ways as shown in [Figure 8.4.1](#). The common support methods are

- **Simply Supported** Supported by a pin on one end and a roller at the other.
- **Cantilevered** Fixed at one end, and unsupported at the other.
- **Overhanging** One or both ends overhang the supports.

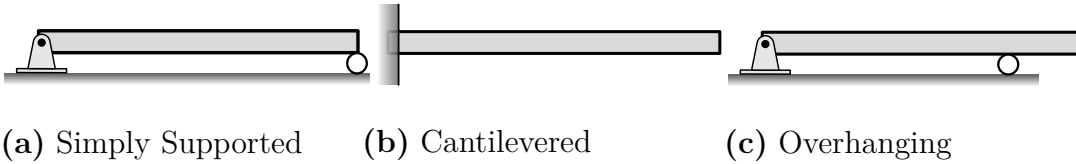


Figure 8.4.1 Beam Supports

We will limit our analysis to these three support systems, and since beams primarily support vertical loads, axial forces will not be considered in this section.

8.5 Relations between Loading, Shear and Moment

Shear and bending moment diagrams are used to show how load on a beam affects the shear and bending moment within.

In this section, we will look at a completely general beam and loading, and use a free-body diagram and the principles of equilibrium to derive the fundamental mathematical relationships that exist between load, shear, and bending moment.

Then in the sections to follow, we will learn three methods to produce shear and bending moment diagrams: an equilibrium based approach, a graphical integration method, and a calculus-based analytical approach. All three methods are consistent with the equations derived in this section and understanding them all can be insightful.

Suppose that we have a simply supported beam upon which there is a load $w(x)$ which acts on the beam by some arbitrary function of x , as shown in Figure 8.5.1.

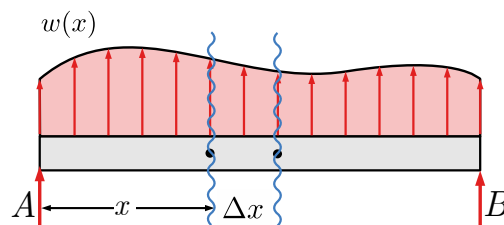


Figure 8.5.1 A simply supported beam with a distributed load that is a function of beam position $w(x)$.

For a thin section of this beam between x and $x + \Delta x$, we can assume that the distributed load is approximately constant and equal to $w = w(x)$. With

this assumption, the distributed load is rectangular, and equivalent to a force $W = w\Delta x$ acting at the centroid of the rectangle. The free-body diagram for this thin section is shown in [Figure 8.5.2](#).

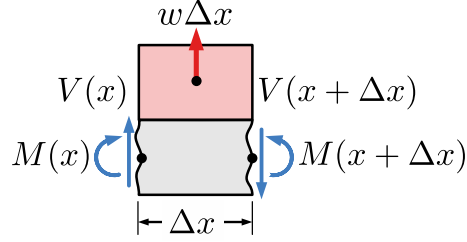


Figure 8.5.2 A free-body diagram of a small section of the beam with a width of Δx .

Applying the force equilibrium in the vertical direction gives the following result:

$$\begin{aligned}\sum F_y &= 0 \\ V + w\Delta x &= (V + \Delta V) \\ \Delta V &= w\Delta x \\ \frac{\Delta V}{\Delta x} &= w\end{aligned}$$

Taking the limit of both sides as Δx approaches 0, we get this important result

$$\begin{aligned}\lim_{\Delta x \rightarrow 0} \left(\frac{\Delta V}{\Delta x} \right) &= \lim_{\Delta x \rightarrow 0} (w) \\ \frac{dV}{dx} &= w(x)\end{aligned}\tag{8.5.1}$$

This differential equation tells us that at a given location x , the slope of the shear function at that point is the value of the loading directly above, $w(x)$.

Furthermore, if we separate the variables by multiplying both sides by dx , and then integrate between any two points along the beam we find that

$$\Delta V = \int_a^b w(x) dx\tag{8.5.2}$$

In words, this equation says that the change in the shear ΔV between any two points a and b , is the area under the loading curve between them.

Next, considering moment equilibrium of the FBD in [Figure 8.5.2](#), and following the same procedure, we get:

$$\begin{aligned}\sum M &= 0 \\ (M + \Delta M) &= \frac{\Delta x}{2}V + \frac{\Delta x}{2}(V + \Delta V) + M\end{aligned}$$

$$\Delta M = \left(V + \frac{\Delta V}{2}\right)\Delta x$$

Separating variables and taking limits as before, gives:

$$\begin{aligned}\lim_{\Delta x \rightarrow 0} \left(\frac{\Delta M}{\Delta x}\right) &= \lim_{\Delta x \rightarrow 0} \left(V + \frac{\Delta V}{2}\right) \\ \frac{dM}{dx} &= V(x)\end{aligned}\tag{8.5.3}$$

This equation tells us that at any point on the beam the slope of the moment function is equal to the value of the shear at that point.

Furthermore, if we multiply both sides by dx , we can integrate to find that

$$\Delta M = \int_a^b V(x) dx\tag{8.5.4}$$

In words, this equation says that over a given segment, the change in the moment value is the area under the shear curve.

Summary. The internal shear force $V(x)$, bending moment $M(x)$, and loading $w(x)$ at every point x are related to each other by the integral and differential equations below.

$$\frac{dV}{dx} = w(x)$$

The slope of the shear function at x is the value of the loading function at the same position. An upward load is considered a positive load.

$$\Delta V = \int_a^b w(x) dx$$

The change in the shear value between two points is the area under the loading function between those points.

$$\frac{dM}{dx} = V(x)$$

The slope of the moment function at x is the value of the shear at the same position.

$$\Delta M = \int_a^b V(x) dx$$

The change in the moment value between two points is the area under the shear curve between those points.

Remark 8.5.3 Sign Conventions.

It is sometimes convenient to assume that positive loads act down, and not up as we have assumed here. In that case simply substitute $-w(x)$ for $w(x)$ into and equations (8.5.1) and (8.5.2) and use these variations instead:

$$\begin{aligned}\frac{dV}{dx} &= -w(x) & \Delta V &= -\int_a^b w(x) dx\end{aligned}\tag{8.5.5}$$

Thinking Deeper 8.5.4 What happens if we keep integrating?

As we have seen, integrating the load function once gives the shear force, and integrating the shear force gives the bending moment.

You may be wondering what happens if we keep integrating? Does this produce anything interesting?

In fact it does, as you will see in Mechanics of Materials. In the Statics world, all beams are assumed to be rigid, but in the real world they deform under load. The shape the beam takes on is described by a deflection curve which gives the actual displacement of the beam at any given point due to the applied loads

Integrating the bending moment $M(x)$ (and dividing by the flexural rigidity, EI) gives the slope of the deflection curve, which represents the angle of the beam's deformation at any given point.

Integrating that function once more (and dividing by EI again) gives the actual deflection function.

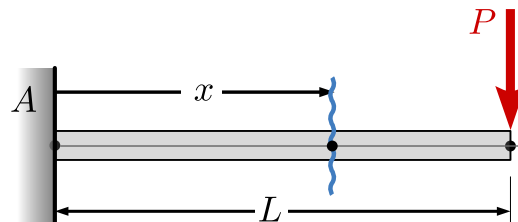
So, if you keep integrating the load function, you're essentially moving through these stages of beam analysis, progressing from the applied forces to the resulting deformation: Load $\rightarrow$ Shear $\rightarrow$ Moment $\rightarrow$ Slope $\rightarrow$ Deflection. Each step provides valuable information about the beam's behavior under the given loading conditions.

8.6 Section Cut Method

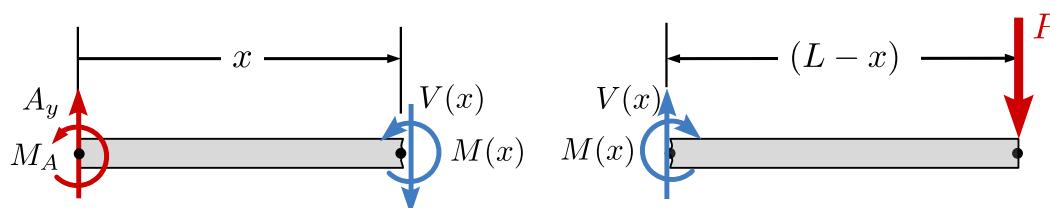
In this section we'll extend the method of [Section 8.3](#) where we found the internal forces at a specific point to find the internal forces at every point needed to produce a shear and bending moment diagram.

The procedure is identical except that the cut is taken at a position designated by x instead of at a specified point. The analysis produces *equations* as functions of x rather than *values* at a particular x . Shear and bending moment diagrams are plots of these equations, and the internal forces at any particular point can be found by substituting that particular value of x into the equations.

We will use a cantilevered beam fixed to a wall on the left end and subjected to a vertical force P on the right end as an example. Global equilibrium requires that the reactions at the fixed support at A are a vertical force $A_y = P$, and a counter-clockwise moment $M_A = PL$.



By taking a cut at a distance x from the left we can draw two free-body diagrams with lengths x and $(L - x)$. This beam has *one* loading segment, because no matter where x is chosen, the free-body diagrams shown in [Figure 8.6.1](#) (b) and (c) are correct. The internal loadings are named $V(x)$ and $M(x)$ to indicate that they are functions of x and are drawn with positive sense as defined in [Section 8.2](#).



(a) Left FBD.

(b) Right FBD.

Figure 8.6.1

To find the shear and bending moment functions, we apply the equilibrium to one of the free-body diagrams. Either side will work, so we'll select the right-hand portion [Figure 8.6.1\(b\)](#) since it doesn't require us to find the reactions at A. Letting L be the length of the beam and $(L - x)$ the length of the right portion, we find

$$\begin{aligned}\Sigma F_y &= 0 & \Sigma M_{\text{cut}} &= 0 \\ V(x) &= P & M(x) &= -P(L - x)\end{aligned}$$

The plots of the equations for $V(x)$ and $M(x)$ are shown below in [Figure 8.6.2](#). These equations indicate that the shear force $V(x)$ is constant P over the length of the beam and the moment $M(x)$ is a linear function of the position of the cut, x starting at $-PL$ at $x = 0$ and linearly increasing to zero at $x = L$. Note that the graphs are only valid from $0 \leq x \leq L$, so the curves outside this range are shown as dotted lines. These two graphs are usually drawn stacked beneath the load diagram of the beam.

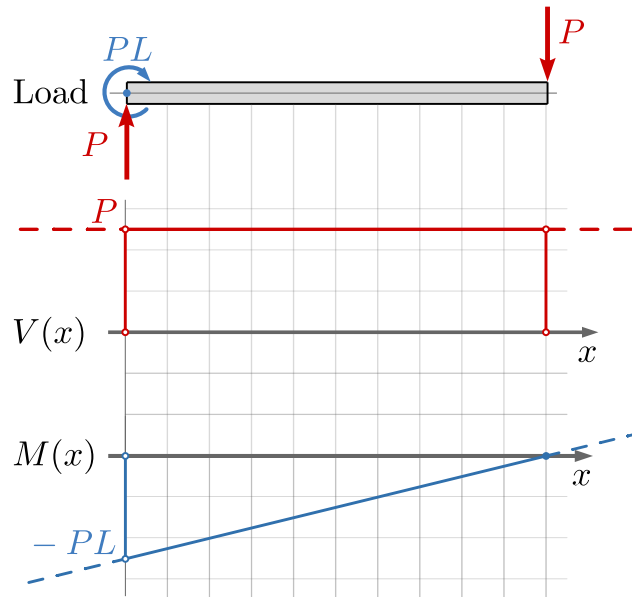


Figure 8.6.2 Shear and Bending Moment Diagrams for Cantilever Beam

The previous example was simple because only one FBD was necessary for any point on the beam, but many beams are more complex. Beams with multiple loads must be divided into loading segments between the points where loads are applied or where distributed loads begin or end.

Consider the simply supported beam AD with a uniformly distributed load w over the first segment from A to B , and two vertical loads B and C .

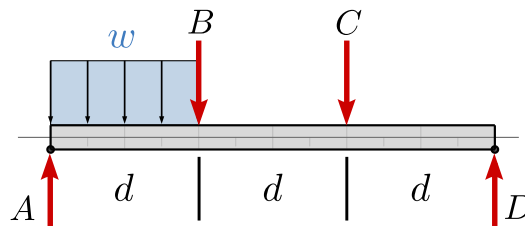
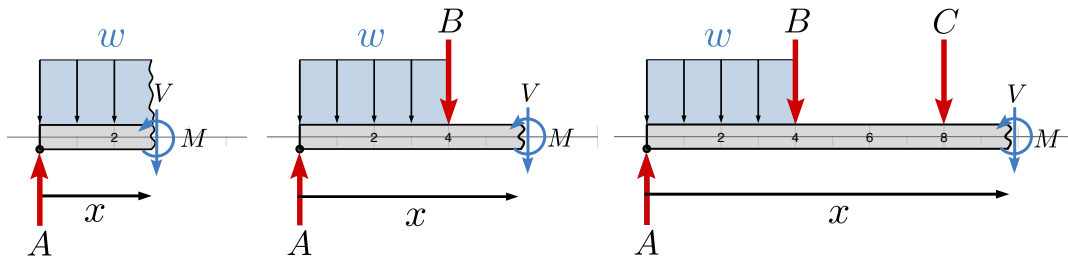


Figure 8.6.3 Simply Supported Beam with Uniformly Distributed Load

This beam has three loading segments so you must draw three free-body diagrams and analyze each segment independently. For each, make an imaginary cut through the segment, then draw a new free-body diagram of the portion to the left (or right) of the cut. Always assume that the exposed internal shear force and internal bending moment act in the positive direction according to the [sign convention](#).



(a) $0 < x < d$ (b) $d < x < 2d$ (c) $2d < x < 3d$

Figure 8.6.4

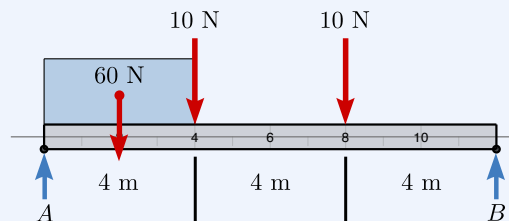
After the equilibrium equations are applied to each segment, the resulting equations $V(x)$ and $M(x)$ from each segment are joined to plot the shear and moment diagrams. These diagrams help us visualize the values of V and M throughout the beam.

Example 8.6.5 Section Cut Method.

Draw the shear and bending moment diagrams for the beam of [Figure 8.6.3](#) knowing that d is 4 m, forces B and C are both 10 N, and the uniformly distributed load $w(x) = 15$ N/m

Solution.

1. Replace the 15 N/m with an equivalent force of 60 N acting at the centroid of the the rectangular loading, and draw a free body diagram of the whole beam.

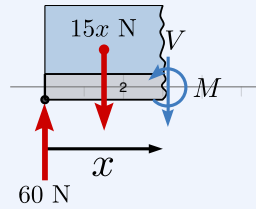


2. Take moments about B to find the reaction at A .

$$A = \frac{60(10) + 10(8) + 10(4)}{12}$$

3. Draw a free-body diagram of the section of the beam between $0 < x < 4$ m.

The loading on this segment of the beam is $w(x) = 15$ N/m, acting down and constant. The equivalent force is the area of the rectangle $W = 15x$, which is a first degree function of x .

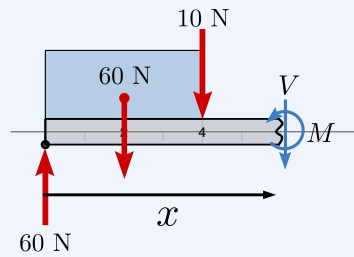


4. Apply the equations of equilibrium to find the shear and moment functions for the segment $0 < x < 4$.

$$\begin{aligned}\Sigma F_y &= 0 \\ 60 - 15x - V &= 0 \\ V &= 60 - 15x \text{ N}\end{aligned}\tag{8.6.1}$$

$$\begin{aligned}\Sigma M_{\text{cut}} &= 0 \\ -60x + (15x)\left(\frac{x}{2}\right) + M &= 0 \\ M &= 60x - 7.5x^2 \text{ N}\cdot\text{m}\end{aligned}\tag{8.6.2}$$

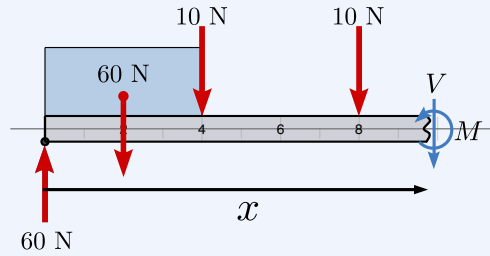
5. Follow a similar procedure for the next segment $4 < x < 8$



$$\Sigma F_y = 0 : \quad V = -10 \text{ N}\tag{8.6.3}$$

$$\Sigma M_{\text{cut}} = 0 : \quad M = 160 - 10x \text{ N}\cdot\text{m}\tag{8.6.4}$$

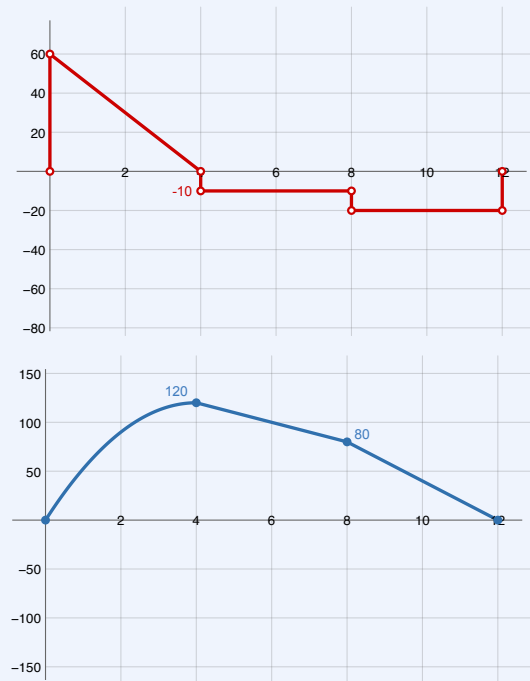
6. And again for the third segment $8 < x < 12$



$$\Sigma F_y = 0 : \quad V = -20 \text{ N} \quad (8.6.5)$$

$$\Sigma M_{\text{cut}} = 0 : \quad M = 240 - 20x \text{ N}\cdot\text{m} \quad (8.6.6)$$

7. Neatly plot these functions on appropriately scaled graph paper



Note that this method does not give us shear and moment values at exactly $x = 4 \text{ m}$ and $x = 8 \text{ m}$. Discontinuities occur at these locations, and the shear function has different values when approaching 4 and 8 from the left and from the right!

8.7 Graphical Method

Once you have a firm grasp on the relations between load, shear and bending moments, you can quickly draw shear and bending moment diagrams using a graphical approach. This technique is really a graphical integration process;

rather than using calculus, the integrals are found using simple geometry.

8.7.1 Relationship Review

Shear and bending moment diagrams are governed by equations (8.5.1) through (8.5.4) and must be consistent with them.

$$\frac{dV}{dx} = w(x)$$
$$\Delta V = \int_a^b w(x) \, dx$$

$$\frac{dM}{dx} = V(x)$$
$$\Delta M = \int_a^b V(x) \, dx$$

From top to bottom, the shear function is determined from the area under the load, and the moment function is determined from the area under the shear function. At the same time, from bottom to top, differentiating the moment function gives the shear function, and differentiating the shear function gives the load.

Shear and bending moment are piecewise functions made up of a series of jumps, slopes and areas. **Jumps** are vertical changes in shear and moment diagrams caused by concentrated forces and moments. **Slopes** are gradual changes in the functions. **Areas** are the ‘areas’ under the loading and shear curves, i.e. the integral. The area under the loading curve is actually a force, and the area under the shear curve is actually a bending moment.

The relationship between load, shear, and bending moment are summarized below.

Shear Diagram	
Jumps	Concentrated forces cause the shear curve to jump by the amount of the force. Upward loads cause upward jumps.
Slopes	The slope of the shear diagram at a point is equal to the value of the distributed load above that point. Downward distributed loads create negative slopes on the shear diagram.
Areas	The area under the loading curve between two points is equal to the corresponding change in value of the shear.
Moment Diagram	
Jumps	Concentrated moments cause the moment curve to jump by the amount of the moment. Counterclockwise moments cause downward jumps.
Slopes	The slope of the moment diagram at a point is equal to the value of the shear at that point. Positive shears cause positive slopes on the moment diagram.
Areas	The area under the shear curve between two points is equal to the corresponding change in the value of the moment.

You can use the interactive below to explore how changes to concentrated load P and distributed load w affect the slopes, jumps, and areas of the resulting shear and bending moment diagrams.

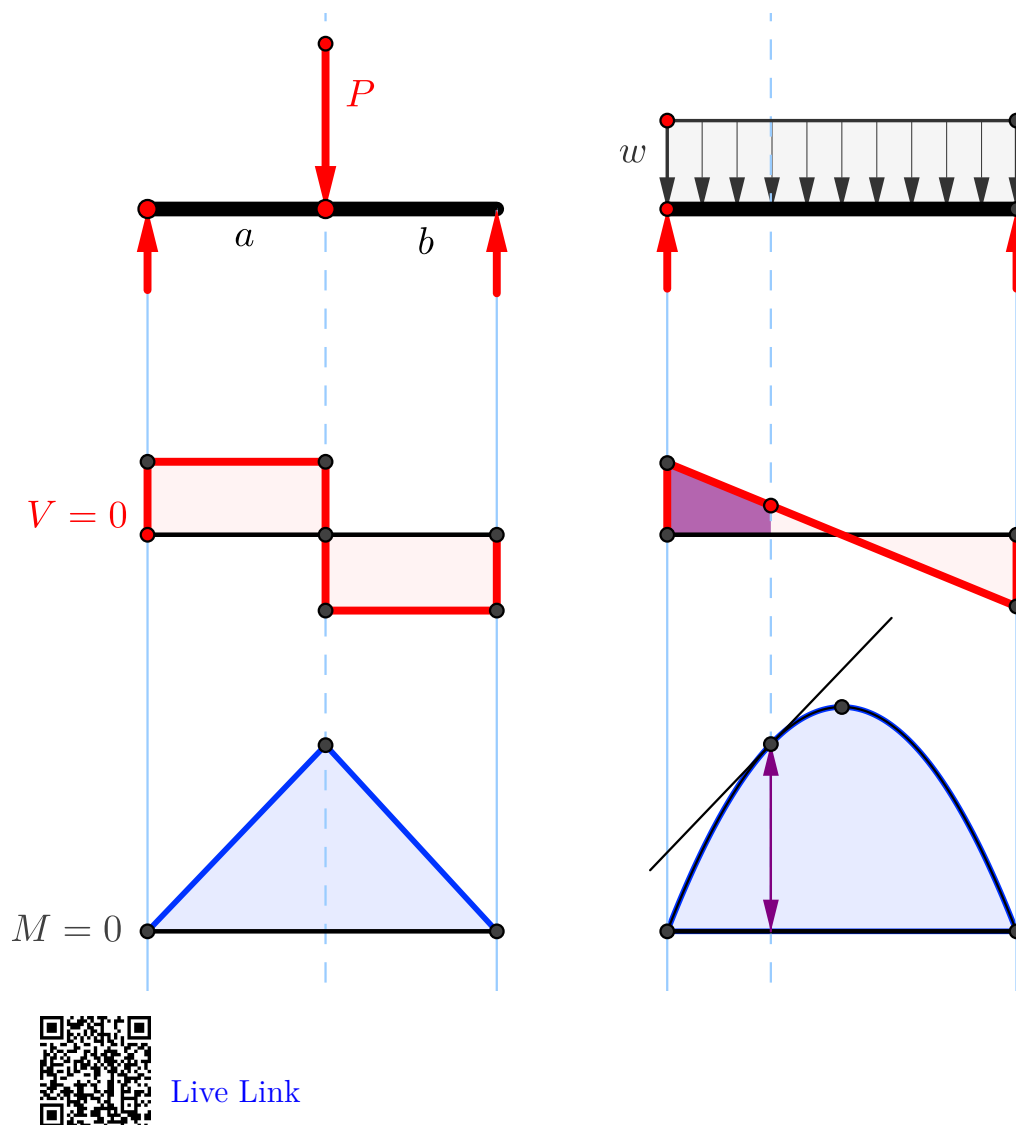


Figure 8.7.1 Building Blocks for Shear and Moment Diagrams

8.7.2 Graphical Method Procedure

You can draw shear and bending moments efficiently and accurately using this procedure

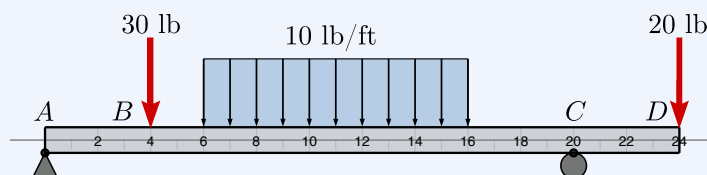
1. First, *determine the reaction forces* and moments by drawing a free-body diagram of the entire beam and applying the equilibrium equations. Double check that your reactions are correct.
2. Establish the shear graph with a horizontal axis below the beam and a vertical axis to represent shear. Positive shears will be plotted above the x axis and negative below.

3. Make vertical lines at all the “interesting points”, i.e. points where concentrated forces or moments act on the beam and at the beginning and end of any distributed loads. This divides the beam into segments between vertical lines.
4. *Draw the shear diagram* by starting with a dot at $x = 0$, $V = 0$ then proceeding from left to right until you reach the end of the beam. Choose and label a scale which keeps the diagram a reasonable size.
 - (a) Whenever you encounter a concentrated force, jump up or down by that value
 - (b) Whenever you encounter a concentrated moment, do not jump.
 - (c) Whenever you encounter a distributed load, move up or down by the “area” under the loading curve over the length of the segment, according to equation (8.5.2). The “area” is actually a force.
 - (d) The slope of the curve at each point x is given by (8.5.1). Distributed loads cause the shear diagram to have a slope equal to value of the distributed load at that point. For unloaded segments of the beam, the slope is zero, i.e. the shear curve is horizontal. For segments with uniformly distributed load, the slope is constant. Downward loads cause downward slopes.
 - (e) The shear diagram should start and end at $V = 0$. If it doesn’t, recheck your work.
5. Add another interesting point wherever the shear diagram crosses the x -axis, and determine the x position of the zero crossing.
6. After you have completed the shear diagram, *calculate the area under the shear curve* for each segment. Areas above the axis are positive, areas below the axis are negative. The areas represent moments and the sum of the areas plus the values of any concentrated moments should add to zero. If they don’t, then recheck your work.
7. Establish the moment graph beneath the shear diagram with a horizontal axis below the shear diagram and a vertical axis to represent moment. Positive moments will be plotted above the x axis and negative below.
8. *Draw and label dots on the moment diagram* by starting with a dot at $x = 0$, $M = 0$ then proceed from left to right placing dots until you reach the end of the beam. As you move over each segment move up or down from the current value by the “area” under the shear curve for that segment and place a dot on the graph. In this step, you are applying (8.5.4).
 - (a) Positive areas cause the moment to increase, negative areas cause it to decrease.

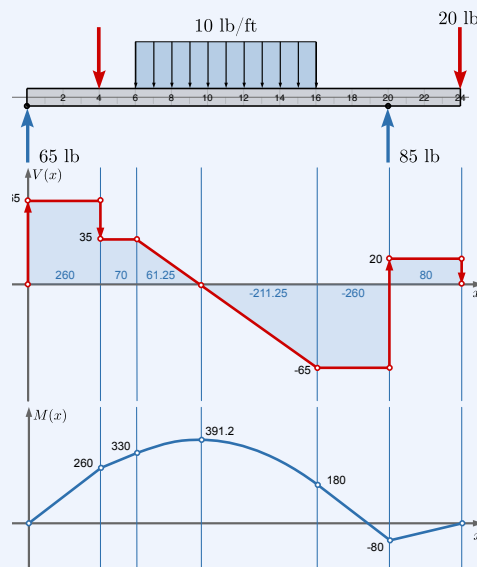
- (b) If you encounter a concentrated moment, jump straight up or down by the amount of the moment and place a dot. Clockwise moments cause upward jumps and counter-clockwise moments cause downward jumps.
- (c) When you reach the end of the beam you should return to $M = 0$. If you don't, then recheck your work.
9. *Connect the dots with correctly shaped lines.* Segments under constant shear are straight lines, segments under changing shear are curves. The general curvature of the lines can be determined by considering equation (8.5.3).

Example 8.7.2 Graphical Method: Combined Loads.

Draw a neat, accurate, labeled shear and bending moment diagram for the beam and loading shown using the method of this section. Distance units are in feet.



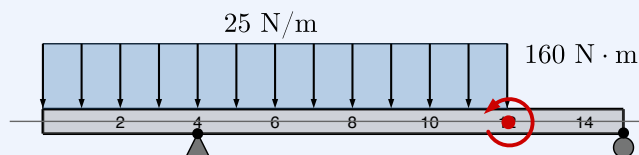
Solution. Click the buttons below to step through the solution.



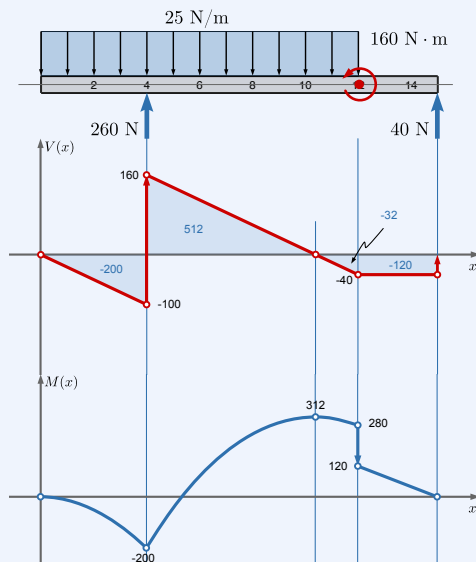
[Standalone](#)
[In Context](#)
[Embed](#)

Example 8.7.3 Graphical Method: Concentrated Moments.

Draw a neat, accurate, labeled shear and bending moment diagram for the 15 m long beam loaded as shown using the method of this section.



Solution. Click the buttons below to step through the solution.



Standalone
In Context
Embed

Remark 8.7.4 Tips for Drawing Good Shear and Bending Moment Diagrams.

1. Work neatly and professionally, and take pride in your work. Use a straightedge!
2. Draw a labeled free-body diagram of the entire structure, and your work to find the reactions off to the side. For this free-body diagram, you may replace the distributed loads with equivalent concentrated loads.
3. Next draw a careful sketch of the beam showing its reactions and the *actual* distributed loads. Distributed loads must be shown here, because their distributed nature is significant.
4. Draw the shear diagram directly below the load diagram and the moment diagram beneath that. It is convenient to draw these graphs on graph paper. Indicate the scale used each function.
5. Downward forces make the shear diagram move down, counterclockwise concentrated moments make the moment curve jump down.

6. Show the correct shape and curvature for each curve segment: zero, constant slope, polynomial. Changes in curve shapes should align with the load which causes them.
7. Indicate the values of shear and moment at maximums, minimums and points of inflection.
8. Include any other work needed to justify your results.

8.8 Integration Method

In [Section 8.5](#) we saw that loading, shear and bending moments are related by integral and differential equations, and in [Subsection 8.7.2](#) used this knowledge to draw shear and bending moment diagram using a graphical approach. This method is straightforward and fast for concentrated forces and uniformly distributed loads, because they produce triangular and rectangular areas that are easy to calculate but it falls short when the areas are more complicated.

For example, a uniformly varying load is a first degree linear function of x ,

$$w(x) = w_0x$$

It integrates to a second degree parabolic shear function, and a third degree cubic moment function. Finding the zero crossings and areas under these curves with geometry is troublesome. More complex loading functions are even more difficult.

When the loading becomes complex it is better to use the fundamental relations [\(8.5.2\)](#) and [\(8.5.4\)](#) to find the shear and bending moment functions directly

$$\Delta V = \int_a^b w(x) \, dx \qquad \Delta M = \int_a^b V(x) \, dx$$

Rather than finding areas and slopes using geometry, we will directly integrate the load function $w(x)$ once to find the shear function $v(x)$, then integrate that result to find the moment function $m(x)$.

These results are the change in shear and moment over a segment; to find the actual shear and moment functions $V(x)$ and $M(x)$ for the entire beam we will need to find initial values for each segment. This is equivalent to applying boundary conditions to find the constant of integration when solving a differential equation. The initial values come from either the final value of the previous segment or from point loads or point moments. Because of the requirement for these segment starting values, no segment can be computed in isolation from the other segments. Physically this means that the shear and moment along a beam are not just due to the loading in one segment, but are related to the loading on the rest of the beam as well.

8.8.1 Determining Loading Functions

Before you can find shear and bending moment functions with integration you must know the equation for the load on each segment of the beam. These equations may be given in the problem statement if you're lucky, or you may have to determine them from a loading diagram.

When determining equations for loading segments, you may choose either *global equations*, where all segments use the same origin, usually at the left end of the beam, or *local equations*, where each segment uses its own origin, usually at the left end of the segment. Often local equations are easier because you can simply use the variable x in your equations as opposed to " $x + \text{constant}$ ", and you do not have to project the y -intercept values back to an axis system which is not adjacent to the segment. See interactive [Figure 8.8.1](#) to explore the difference between local and global equations.

When determining equations for the loading segments from the load diagram, consider the following.

- *No load.*

Whenever there is no load at all on a segment there will be no change in the shear on the segment. On such sections the loading function is

$$w(x) = 0.$$

Note that this can only occur when the weight of the beam itself is neglected.

- *Point Load.*

A point load is a concentrated force acting at a single point which causes a jump in the shear diagram.

- *Uniformly Distributed Load.*

A uniformly distributed load is constant over the segment and results in a linear slope, either a triangle or a trapezoid, on the shear diagram. The loading function on such sections is

$$w(x) = C \quad V(x) = Cx + b.$$

The constant value is negative if the load points down, and positive if it points upward.

- *Uniformly Varying Load.*

In this case the loading function is a straight, sloping line forming a triangle or trapezoidal shape. The resulting shear function is parabolic. The general form of these functions are

$$w(x) = mx + b \quad V(x) = \frac{mx^2}{2} + bx + c.$$

The slope m , intercept b , and constant c must be determined from the situation, and will depend on whether you are writing a global or local equation.

- *Arbitrary Load.*

The loading function will be a given function of x .

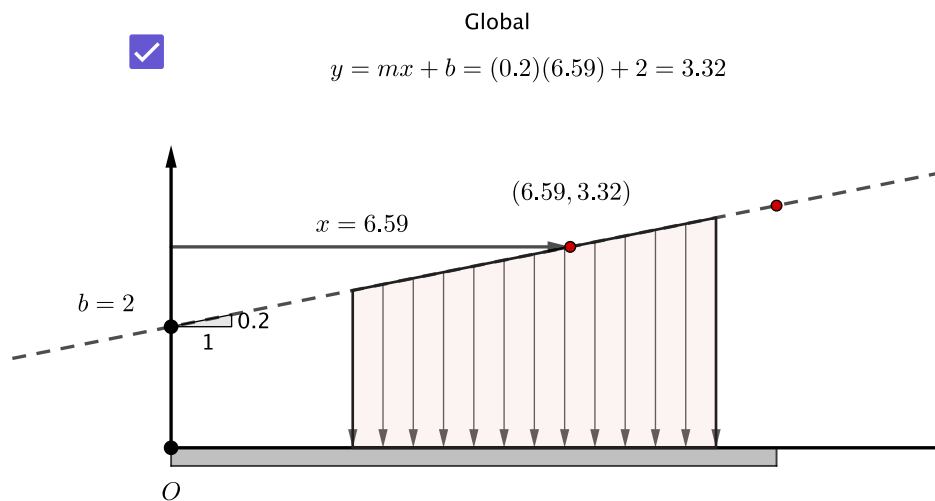
$$w(x) = f(x),$$

and the shear and moment functions are found by integration.

$$V(x) = \int f(x)dx \quad M(x) = \int V(x)dx$$

Most gravitational distributed loads are drawn with the arrows pointing down and resting on the beam. If you slide these along their line of action so that their tails are on the beam, the tips define the loading equation. In other words, for downward pointing loads, the scalar values are negative.

This interactive compares the local and global equations for a beam segment with a uniformly varying load.



[Live Link](#)

Figure 8.8.1 Global vs. Local coordinate systems.

8.8.2 Application of the Integration Method

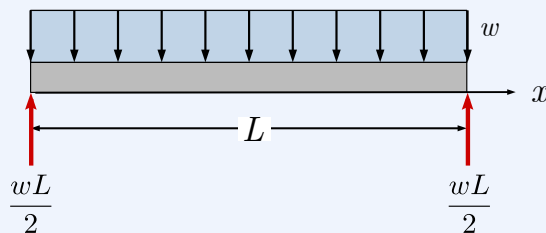
You can either use this method from the start or use the graphical method until you need areas of shapes more complicated than rectangles and triangles.

1. You will need to have solved the loading segment to the left of your desired segment.
2. Write an equation for the loading w in the segment using either local or global coordinates.
3. Integrate the loading equation $w(x)$ to find the change in the shear ΔV and include the shear value at the beginning of your loading segment including the influence of any point loads at that location, which is equivalent to the integration constant.
4. Integrate the shear equation $V(x)$ to find the change in the bending moment ΔM and include the moment value at the beginning of your loading segment including the influence of any point couple-moments at that location, equivalent to the integration constant.
5. To find maximum shear and bending moments, recall from calculus that the local maximum/minimum points of a function occur at the endpoints and where the function's first derivative is equal to zero.
 - (a) For shear, evaluate the shear function $V(x)$ at the ends and where ever the load function crosses the x axis.
 - (b) For bending moments, find the roots of the shear function by solving $V(x) = 0$, then evaluate the moment function $M(x)$ at these points, and also at the endpoints.

The critical values we are looking for are the points where the *magnitudes* of the shear and bending moment are maximum. The direction of the internal forces is not usually significant.

Example 8.8.2 Uniformly Distributed Load.

Use the integration method to find the equations for shear and moment as a function of x , for a simply supported beam carrying a uniformly distributed load w over its entire length L .



Solution.

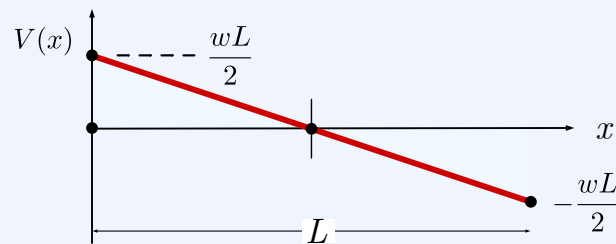
This beam has only one load section, and on that section the load is constant and points down, so,

$$w(x) = -w.$$

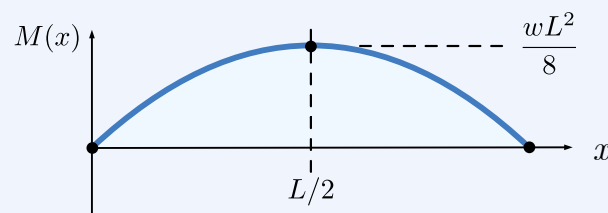
There is a pinned connection at $x = 0$ which provides a vertical force and no concentrated moment, so the initial conditions there are $V(0) = wL/2$, and $M(0) = 0$.

Integrating equations (8.5.2) and (8.5.4) we have.

$$\begin{aligned}\Delta V &= - \int_0^x w(x) \, dx \\ V(x) - \cancel{V(0)}^{wL/2} &= -wx \\ V(x) &= \frac{wL}{2} - wx \\ &= w \left(\frac{L}{2} - x \right)\end{aligned}$$

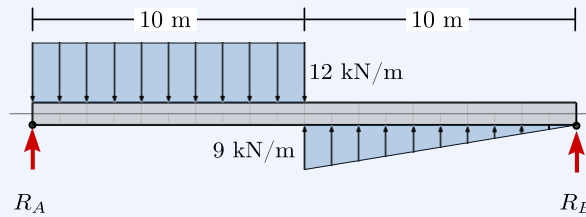


$$\begin{aligned}\Delta M &= \int_0^x V(x) \, dx \\ M(x) - \cancel{M(0)}^0 &= \int_0^x w \left(\frac{L}{2} - x \right) \, dx \\ M(x) &= \frac{w}{2} (Lx - x^2)\end{aligned}$$

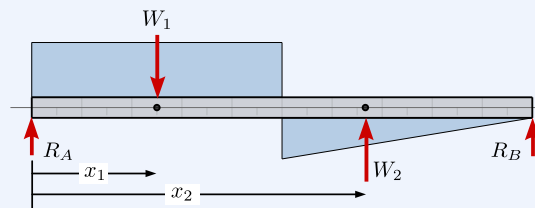


Example 8.8.3 Piecewise Function.

Consider a 20 m beam loaded as shown. Use the integration method to derive the shear and bending moment as functions of x and plot them.

**Solution.**1. *Reactions.*

Find the reactions using the methods of [Section 7.8](#).



$$\Sigma M_A = 0$$

$$-W_1 x_1 + W_2 x_2 + R_B (20 \text{ m}) = 0$$

$$R_B (20 \text{ m}) = 120 \text{ kN} \cdot 5 \text{ m} - 45 \text{ kN} \cdot 13.33 \text{ m}$$

$$R_B = 0$$

$$\Sigma F_y = 0$$

$$R_A - W_1 + W_2 + R_B = 0$$

$$R_A = 120 \text{ kN} - 45 \text{ kN}$$

$$R_A = 75 \text{ kN}$$

2. *Segment 1.*

On the first half of the beam, for the range $x = 0$ up to and including $x = 10$ m, the loading is constant and pointing down, so the loading function on this segment is

$$w_1(x) = -12 \text{ kN/m.}$$

Integrating twice gives the shear and bending moment functions on this segment

$$\begin{aligned}V_1(x) &= -12x + C_0 \\M_1(x) &= -6x^2 + C_0x + C_1\end{aligned}$$

Evaluating the boundary conditions, $V(0) = R_A$ and $M(0) = 0$ we learn that

$$\begin{aligned}C_0 &= 75 \text{ kN.} \\C_1 &= 0\end{aligned}$$

3. Segment 2.

To simplify the analysis of the second half of the beams we will use local coordinates by defining

$$x' = (x - 10).$$

In local coordinates, the function is valid over the range $0 < x' \leq 10$.

The load over this segment starts at 9 kN/m but decreases to zero after 10 m, so the slope is

$$m = \frac{-9 \text{ kN/m}}{10 \text{ m}} = -\frac{9}{10} \text{ kN/m}^2.$$

In terms of x' the load the loading function is

$$w_2(x') = 9 - \frac{9}{10}x'.$$

Integrating twice gives the shear and moment functions.

$$\begin{aligned}V_2(x') &= 9x' - \frac{9}{20}x'^2 + C_2 \\M_2(x') &= \frac{9}{2}x'^2 - \frac{3}{20}x'^3 + C_2x' + C_3\end{aligned}$$

The constants are evaluated by matching the functions at the boundary $x = 10$ m, where

$$\begin{aligned}V_1(10) &= -45 \text{ kN} \\M_1(10) &= 150 \text{ kN}\cdot\text{m}\end{aligned}$$

Evaluating $V_2(x')$ and $M_2(x')$ at $x' = 0$ we find that

$$\begin{aligned}C_2 &= -45 \text{ kN} \\C_3 &= 150 \text{ kN}\cdot\text{m}\end{aligned}$$

4. *Results.*

Combining results we get the final version of shear and bending moment functions in terms of x .

On the range $0 < x \leq 10$

$$V(x) = -12x + 75$$

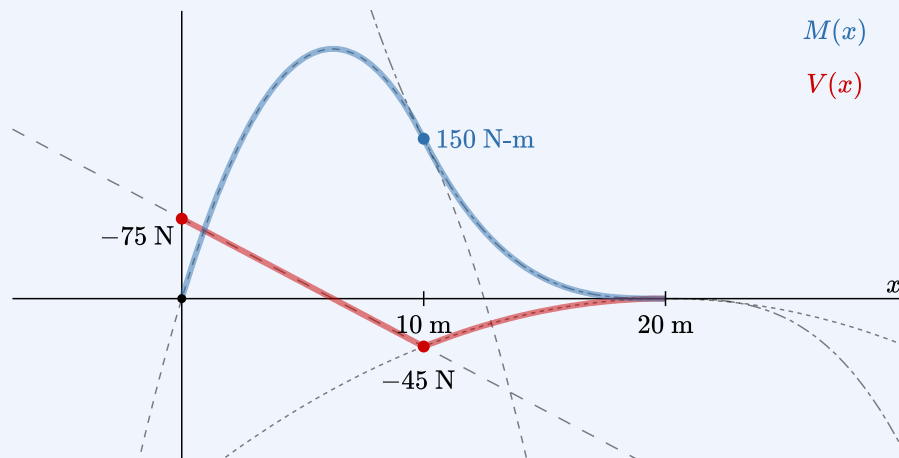
$$M(x) = -6x^2 + 75x$$

On the range $10 < x \leq 20$

$$V(x) = -\frac{9}{20}(x - 10)^2 + 9(x - 10) - 45$$

$$M(x) = -\frac{3}{20}(x - 10)^3 + \frac{9}{2}(x - 10)^2 - 45(10 - x) + 150$$

A plot of the two functions is shown below. Notice how the two halves of each function joins continuously at $x = 10$ m.

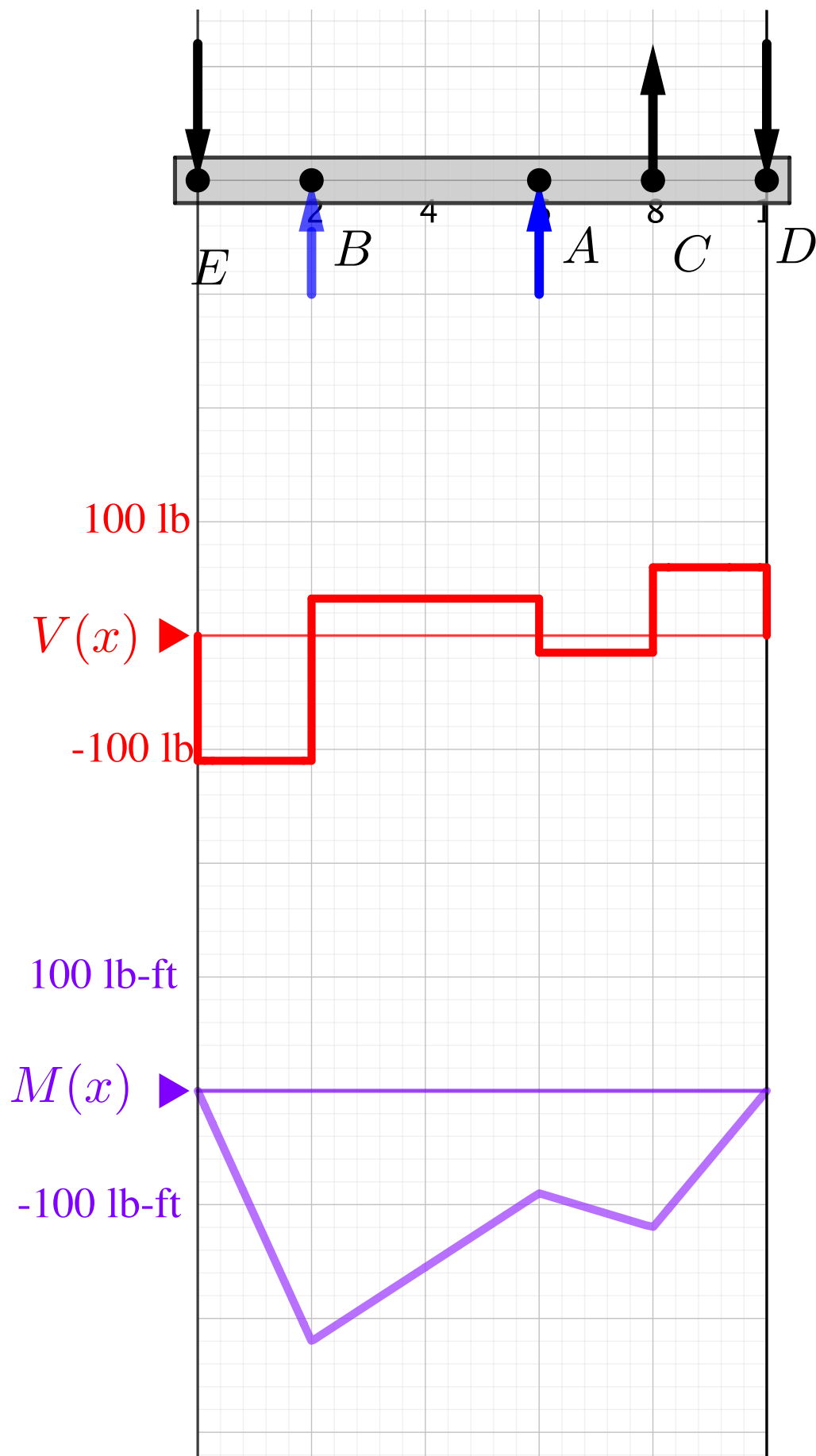


8.9 Geogebra Interactives

You can use the interactives below to practice drawing shear and bending moment diagrams. You can change the position of the loads and change their magnitudes and observe how the diagrams change, or you can click **New Problem** to generate a new problem.

After exploring the diagrams and seeing how they relate to each other, turn off the solutions and try for yourself. You will very likely see problems like these on an exam.

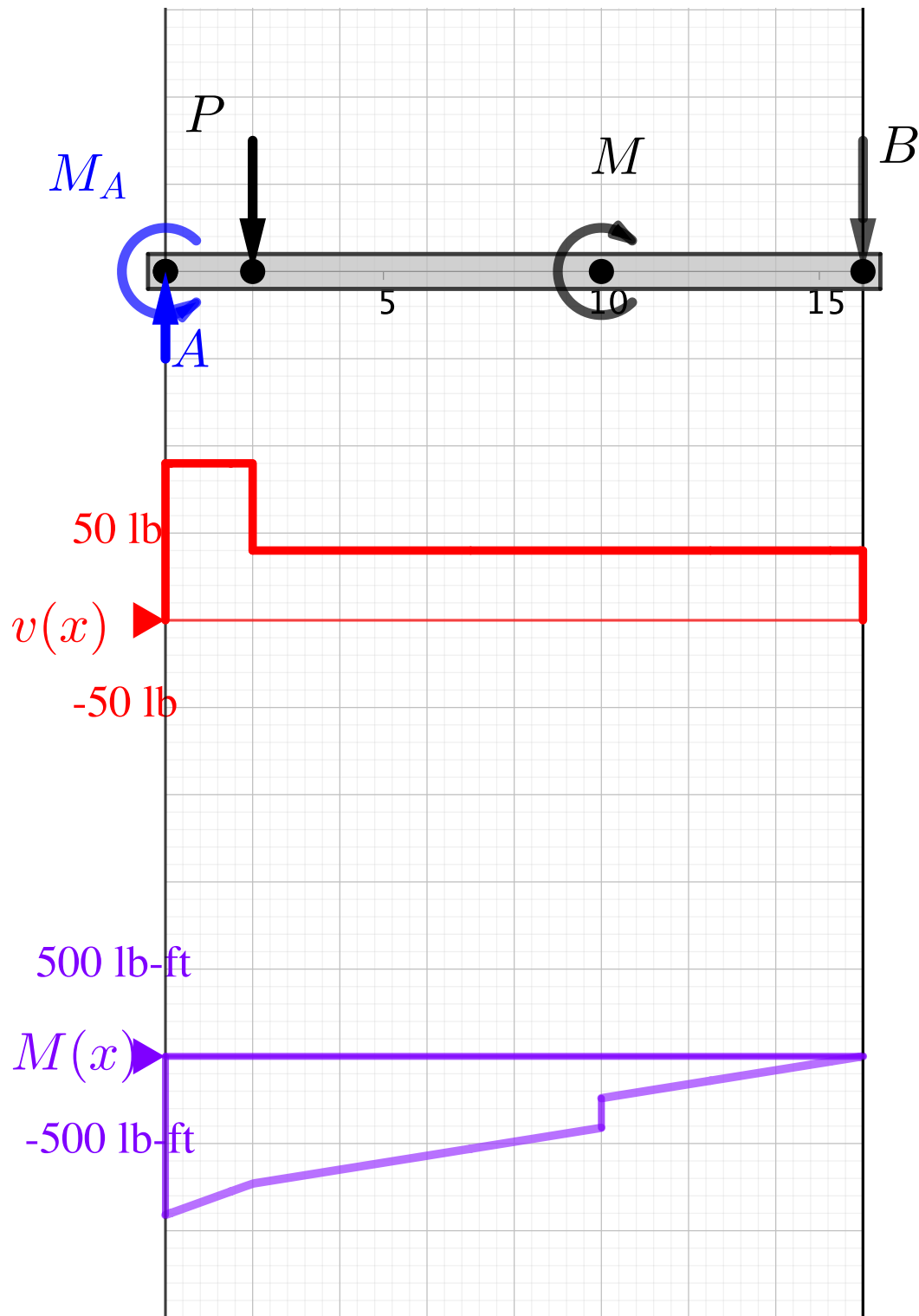
8.9.1 Concentrated Forces





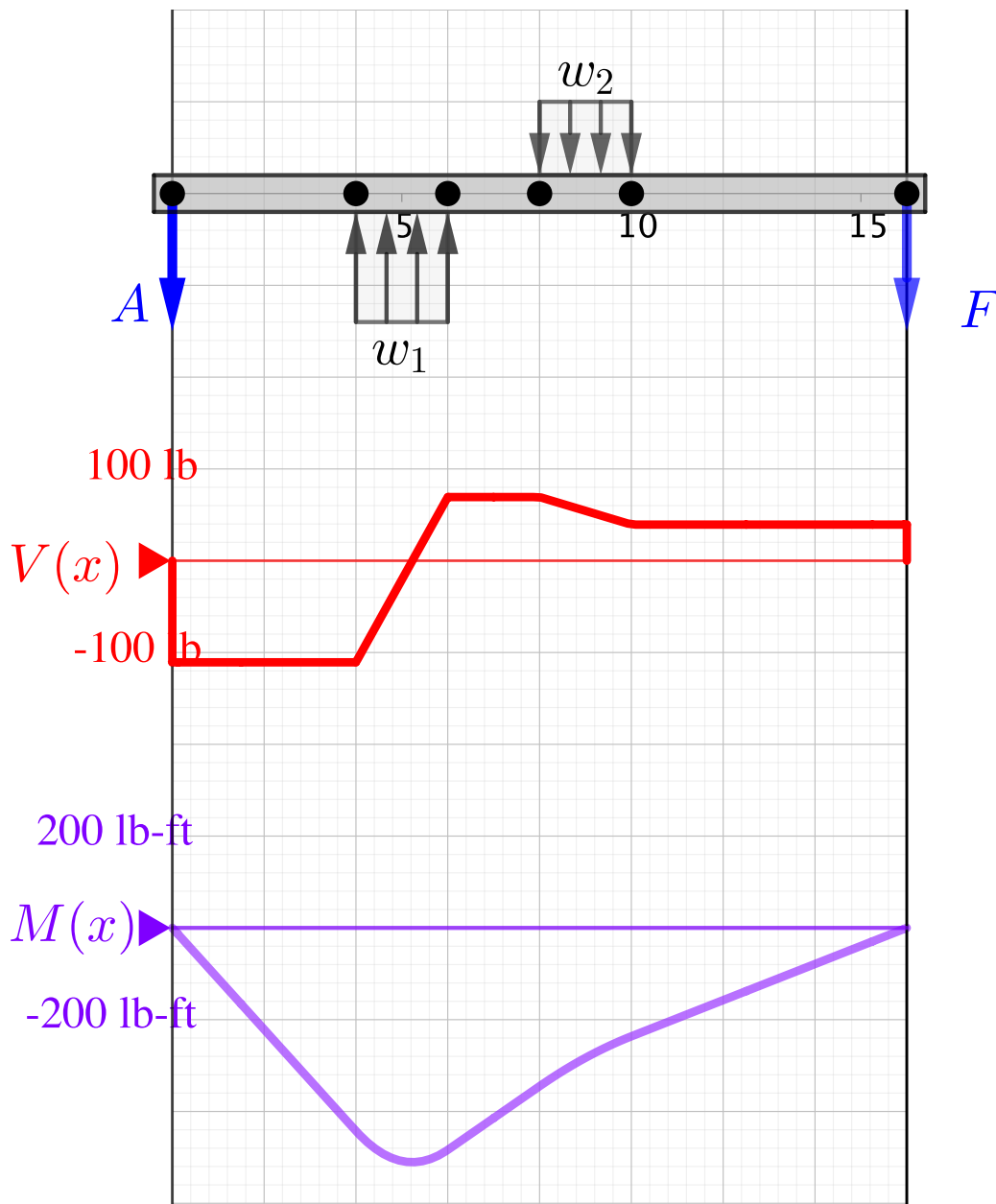
[Live Link](#)

8.9.2 Concentrated Force and Moment

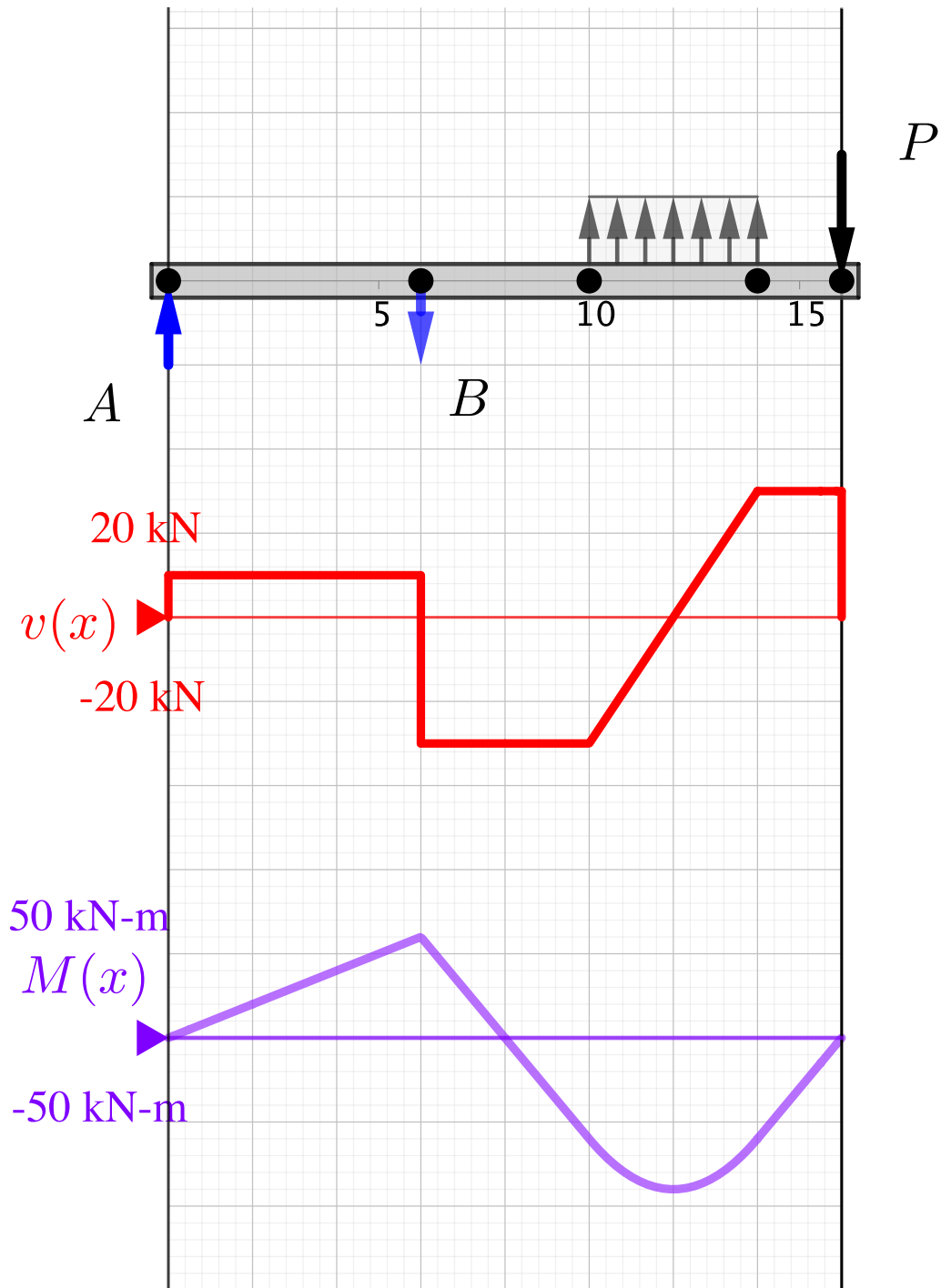


[Live Link](#)

8.9.3 Distributed Load

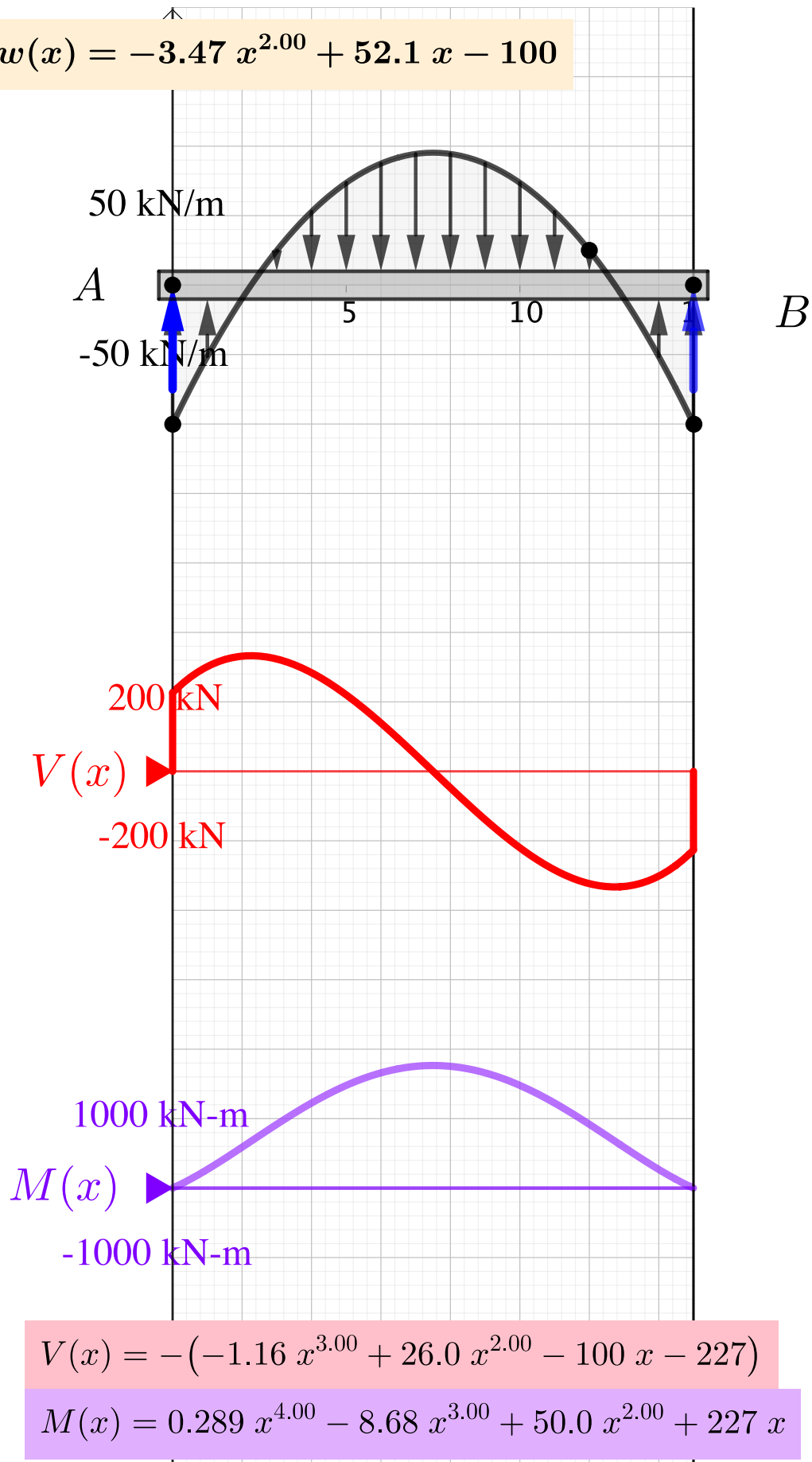
[Live Link](#)

8.9.4 Combination Load

[Live Link](#)

8.9.5 Arbitrary Load

$$w(x) = -3.47 x^{2.00} + 52.1 x - 100$$



[Live Link](#)

8.10 Summary

You have likely realized that in engineering (and life) that there are multiple ways to solve a problem. The four different techniques to compute internal forces discussed in this chapter are a demonstration of this. In the end, the choice of which method to use is yours; the better you know each method the easier it will be to choose the one which is most applicable and efficient.

The list below summarizes each of the four approaches and their advantages and disadvantages.

List 8.10.1 Summary of Methods to find Internal Loadings.

- In [Section 8.3](#) we exposed the internal forces at a specific point inside a rigid body by taking a cut at that location and applying the equilibrium equations. This approach is computationally efficient, works for any rigid body and takes advantage of tools you have learned in previous chapters. It requires knowledge of sign conventions for internal shear and bending moments, but only reveals values at the selected point.
- In [Section 8.6](#) we generalized the previous approach by taking a cut at a variable location, and analyzed the equilibrium equations in terms of x . The results were *functions* which describe shear and moment at every point within the beam, rather than at a specific location. This method requires breaking a beam into loading segments and writing and solving equations for each segment. The equations are then plotted to give shear and bending moment diagrams.
- In [Section 8.7](#) we used the mathematical relationships between loading, shear, and bending moment to draw shear and moment diagrams directly. The method is quick and requires only a few simple computations to determine the critical values. The approach provides many cross-checks for accuracy. It is most suitable for beams loaded with concentrated forces, concentrated moments, and uniformly distributed loads, but is not usually suitable for more complex distributed loads. It is essential to have a solid grasp of the integral and differential relations between loading, shear and moment discussed in [Section 8.5](#) to use this method.
- In [Section 8.8](#) we discussed the most general approach to determine internal loads. In the approach, the load is described as a piece-

wise function of x , which is integrated twice to develop equations for shear and moment. This method can be used for arbitrarily complicated loading distributions, and can be used by software solutions. Applying this method by hand requires accurate integration and differentiation, and application of boundary conditions.

8.11 Exercises (Ch. 8)

Chapter 8 Exercises

Internal Forces	0/80
Distributed load V, M	0/50
	Not attempted
Distributed load V, P, M	0/30
	Not attempted
Shear and Bending Moment Diagrams	0/275
V&M Diagrams: Section Cut Method	0/60
	Not attempted
V&M Diagrams: Concentrated Forces	0/20
	Not attempted
V&M Diagrams: Concentrated Moment	0/20
	Not attempted
V&M Diagrams: Distributed Loads	0/20
	Not attempted
V&M Diagrams: Combined Loads	0/20
	Not attempted
V&M Diagrams: Ground Beam	0/15
	Not attempted
V&M Diagrams: Triangular Loading	0/20
	Not attempted
V&M Diagrams: Symmetric Parabolic Loading	0/60
	Not attempted
V&M Diagrams: Arbitrary Loading Function	0/40
	Not attempted

Display options

Done

Created using Numbas v10.0, developed by Newcastle University.



Standalone
In Context